

Lab 1 – Version 1

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CS 411W

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1 Introduction

The beverage manufacturing industry depends on coordination among brands, suppliers, co-packers, distributors, production personnel, logistics providers, and clients. Many organizations manage these activities through spreadsheets, email chains, phone calls, and disconnected software applications. As beverage companies grow, these fragmented processes can cause communication delays, approval bottlenecks, scheduling conflicts, document-version errors, and limited visibility into production progress. The scale of the industry makes effective coordination increasingly important. The beverage packaging market was valued at approximately \$157.7 billion in 2023 and is projected to reach \$222.1 billion by 2030 (Grand View Research, 2024). The complexity of beverage manufacturing and packaging investments further demonstrates the need for coordinated and scalable information systems (Zimmerman, 2026).

An effective solution must provide a centralized environment in which authorized stakeholders can coordinate production-related activities and access current information. Needed solution characteristics include centralized workflow management, real-time production visibility, automated approval tracking, production scheduling, logistics coordination, document management, reporting and dashboard capabilities, secure role-based access, and scalability. Together, these characteristics would reduce dependence on scattered spreadsheets and messages while helping stakeholders recognize delays, locate current documents, monitor responsibilities, and make informed operational decisions.

BevOS is a centralized workflow-management platform designed specifically for beverage manufacturing operations. The platform connects brands, suppliers, distributors, co-packers, logistics providers, and internal operations through a shared system. BevOS is intended to reduce repetitive manual coordination, improve communication, organize operational information, and increase visibility throughout the production lifecycle without attempting to replace every existing enterprise system or human decision-making process.

A functional prototype will demonstrate how the BevOS user interface, core services, functional modules, document storage, and database work together to address the identified coordination problem. The prototype will illustrate essential workflows involving product requests, quotes, approvals, production activities, logistics, communications, documents, and reporting. This approach will demonstrate the feasibility of the proposed solution and show how its major characteristics support a more coordinated beverage-manufacturing process.

2 BevOS Product Description

Purpose. BevOS is intended to centralize beverage-manufacturing workflows that are otherwise distributed across spreadsheets, email, phone calls, and separate applications. By placing production information and coordination activities in a shared environment, the platform is designed to reduce dependence on manual communication, improve coordination among stakeholders, and support more efficient production management.

Goals and Objectives. The product goals are to improve operational efficiency, reduce communication delays, increase visibility into production activity, and improve the management of approvals. BevOS also aims to support business growth by organizing workflows and

information in a system that can scale as the number of users, projects, and operational activities increases.

Intended Users. BevOS is intended for the organizations and personnel that participate in beverage production and delivery. These users include beverage brands, suppliers, distributors, co-packers, production teams, logistics providers, and finance personnel. Role-based access will allow authorized users to work with the information and functions appropriate to their responsibilities.

2.1 Key Product Features and Capabilities

BevOS will provide centralized workflow and product-request management. Users will be able to create and monitor product requests, organize related work in one platform, and track projects through the production lifecycle. Quote management will record supplier quote requests and responses, while approval tracking will support artwork and purchase-order approvals and make the status of those approvals visible within the workflow.

Production and logistics capabilities will coordinate manufacturing schedules, track production progress, monitor shipping activity, and record delivery status. Document management will store and organize operational files for authorized users. A shared communication capability will keep workflow-related communication with the relevant work, reducing reliance on separate email chains.

Reporting and dashboard capabilities will present operational information so users and managers can see current activity and support decision-making. Issue tracking will record workflow problems so they can be followed and used to support continuous process improvement. Together, these capabilities provide the coordination and visibility identified as necessary for the BevOS solution. The significance of BevOS is its integration of these functions within a single beverage-manufacturing workflow, allowing operational information, communication, approvals, production activity, and logistics information to remain connected throughout the production lifecycle.

2.2 Major Components (Hardware/Software)

Hardware Components. Users will access BevOS from desktop or laptop computers and from supported mobile devices or tablets through an internet connection. The system will rely on cloud-hosted server infrastructure to make the web-based platform and its data available to authorized users.

Software Components. The BevOS software will include a web user interface, a Core Services layer, a PostgreSQL database, and functional modules for reporting, logistics, production, quote management, communications, and document storage. These components divide the platform into user-facing, processing, data-storage, and workflow-specific responsibilities.

System Architecture. Users will interact with BevOS through the web interface. Requests from the interface will be processed by the Core Services layer, which will coordinate the functional modules. The modules will manage production, logistics, quotes, communications, reporting, and

document storage, while operational data will be maintained in PostgreSQL. This organization follows the team's Major Functional Component Diagram (MFCD), shown in Figure 1, and provides the basis for the functional prototype.

As represented in the team's MFCD, brands, distributors, and suppliers interact with BevOS through the web user interface. The interface sends requests to BevOS Core Services, which coordinates the production, logistics, quotes, communications, reporting, and document-storage modules. These modules use PostgreSQL to store and retrieve the structured information required to perform their functions. This separation of the user interface, shared application logic, functional modules, and persistent data supports maintainability and scalability as BevOS evolves.

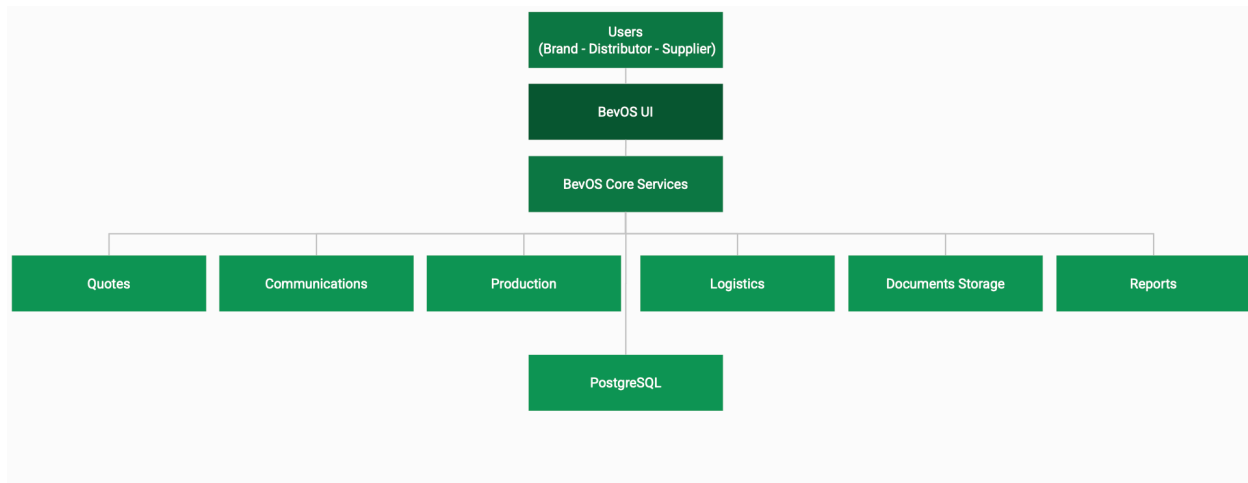


Figure 1 BevOS Major Functional Components Diagram (MFCD)

3 Glossary

- **Approval Workflow** - A process used to review and approve operational tasks before they move to the next stage.
- **BevOS** - A centralized workflow-management platform designed for beverage manufacturing operations.
- **Client** - An organization or individual purchasing services or products from another participant in the manufacturing process.
- **Co-Packer** - A third-party company that manufactures and packages beverages for another company.
- **Core Services** - The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.
- **Dashboard** - A visual interface displaying operational information and performance metrics.
- **Distributor** - An organization responsible for delivering products from manufacturers to retailers or customers.
- **ERP (Enterprise Resource Planning)** - Software that integrates core business processes into a unified system.
- **Logistics** - The planning and coordination of transportation and delivery activities.
- **MFCD (Major Functional Components Diagram)** - A diagram illustrating the primary hardware and software components of the system.
- **Nice Package Company** - The beverage-industry client and domain partner for whom BevOS is being developed.
- **PostgreSQL** - An open-source relational database management system used to store application data.
- **Role-Based Access Control** - A security method that grants system permissions according to the user's assigned role and responsibilities.
- **Single Source of Truth** - A trusted, centralized set of current information used consistently by authorized stakeholders.
- **Stakeholder** - A person or organization that participates in or is affected by a beverage-manufacturing workflow.
- **Workflow** - A sequence of business activities completed to achieve a specific objective.

4 References

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